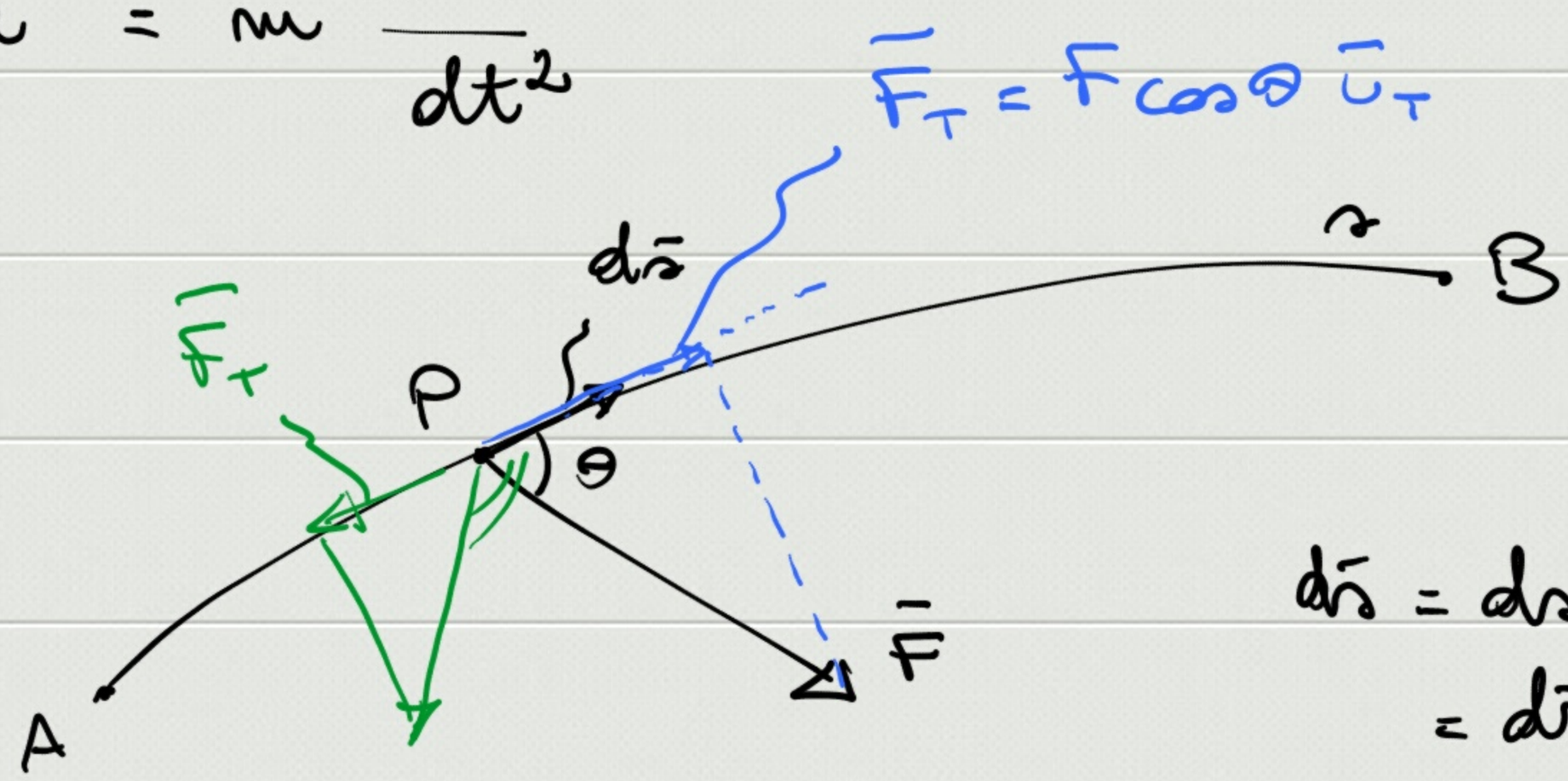
$$\bar{F} = m\bar{a} = m\frac{d^2\bar{r}}{dt^2}$$



$$d\bar{s} = ds\,\bar{u}_T = d\bar{r}$$

Lavoro (infinitesimo)

$$\boxed{dW = \bar{F}\cdot d\bar{s}}$$

$$dW = \bar{F}\,d\bar{s} = F\,ds\cos\theta = F_T\,ds$$

$$m\frac{dv}{dt}$$

$0 \le \theta < \frac{\pi}{2} \Rightarrow \cos\theta > 0 \Rightarrow dW > 0$ lavoro motore

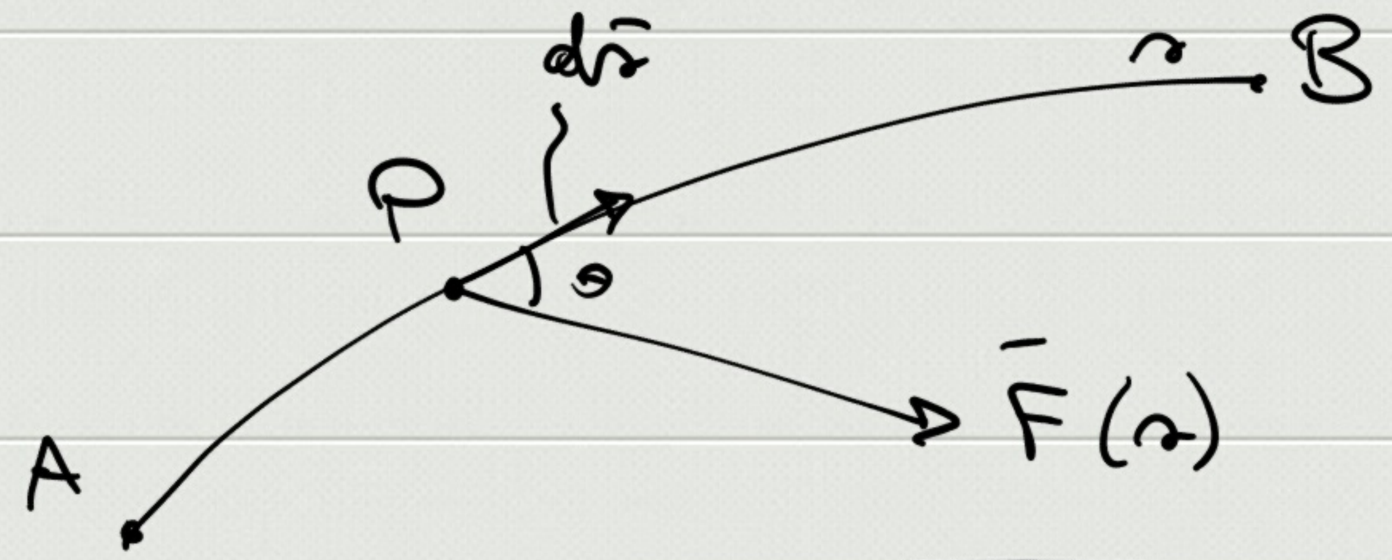
$v \uparrow$

$\frac{\pi}{2} < \theta \le \pi \Rightarrow \cos\theta < 0 \Rightarrow dW < 0$ lavoro resistente

$v \downarrow$

$\theta = \frac{\pi}{2}$ $(\bar{F} \perp d\bar{s}) \Rightarrow \bar{F} = \bar{F}_N$ $(F_T = 0)$

$\cos\theta = 0 \Rightarrow dW = 0$ (lavoro nullo)



$$\boxed{W_{A\to B} = \int_A^B dW = \int_A^B \bar{F}\, d\bar{s} = \int_A^B F_T\, ds}$$

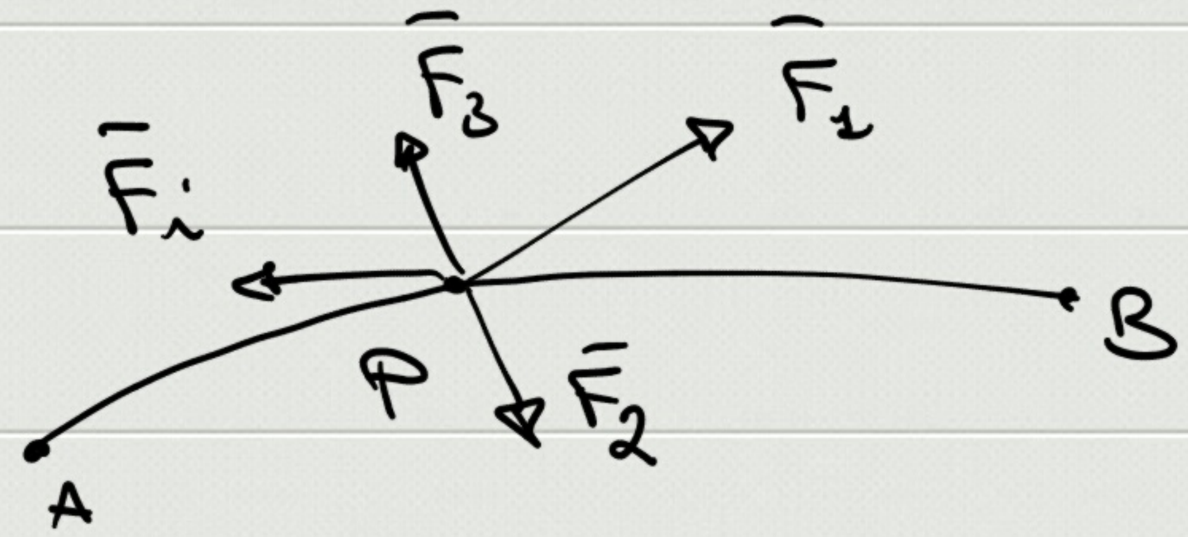
$$d\bar{s} = dx\,\bar{u}_x + dy\,\bar{u}_y + dz\,\bar{u}_z$$

$$\bar{F} = F_x\,\bar{u}_x + F_y\,\bar{u}_y + F_z\,\bar{u}_z$$

$$\Rightarrow W_{A\to B} = \int_A^B \bar{F}\, d\bar{s} = \int_A^B \left(F_x\,dx + F_y\,dy + F_z\,dz\right) =$$

$$= \int_{x_A}^{x_B} F_x\,dx + \int_{y_A}^{y_B} F_y\,dy + \int_{z_A}^{z_B} F_z\,dz$$

$$W_{A\to B} = \int_A^B dW \cancel{= W_B - W_A}$$

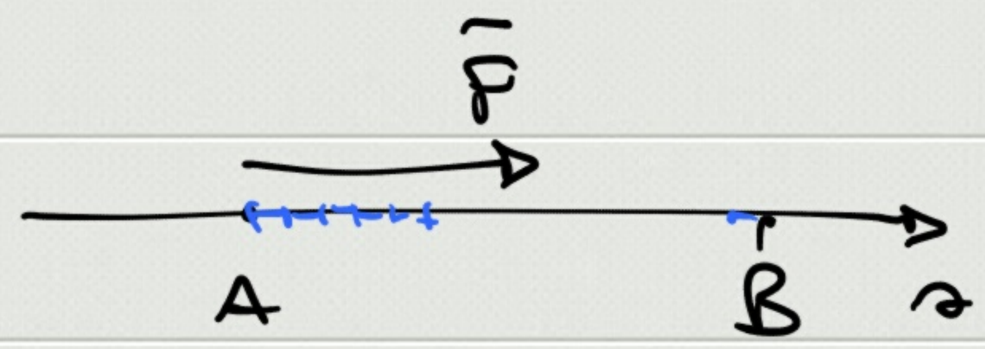


$$\bar{R} = \sum_i \bar{F}_i$$

$$W_{A\to B} = \int_A^B dW = \int_A^B \bar{R}\, d\bar{s} = \int_A^B \sum_i \bar{F}_i\, d\bar{s} =$$

$$= \sum_i \underbrace{\int_A^B \bar{F}_i\, d\bar{s}} = \sum_i \left(W_{A\to B}\right)_i$$

$$\boxed{\bar{F} = \text{cost} \;/\!/\; d\bar{s}}$$



$$W_{A\to B} = \int_A^B \bar{F}\, d\bar{s} = \int_A^B F\, ds = F \int_A^B ds = F\,|\bar{AB}|$$

Potenza : lavoro per unità di tempo

$$dW = \bar{F}\, d\bar{s}$$

$$\boxed{P = \frac{dW}{dt}}$$

$$P = \frac{dW}{dt} = \bar{F}\,\frac{d\bar{s}}{dt} = \bar{F}\cdot\frac{d\bar{s}}{dt} = \bar{F}\cdot\bar{v}$$

$$\langle P \rangle_{\Delta t} = \frac{W_{\Delta t}}{\Delta t}$$

$$[W] = [F\cdot s] = N\,m = kg\,m^2 s^{-2} = J$$

(Joule)

$$[P] = \left[\frac{W}{t}\right] = \frac{J}{s} = kg\,m^2 s^{-3} = W \quad \text{(Watt)}$$